

Sanjit Sai Kolla

Associate Machine Learning Engineer

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SUMMARY

Applied Machine Learning Engineer with ~2 years of experience building production-grade AI systems in the financial domain. Specializes in LLM-powered automation, Retrieval-Augmented Generation (RAG), and model fine-tuning. Strong foundation in Python, SQL, and modern LLM frameworks. U.S. citizen — no visa sponsorship required.

EXPERIENCE

Associate Machine Learning Engineer

June 2024 - Present

Jocata Financial Advisory & Technology | Hyderabad, India

AI-Assisted Support Bot

- Built a production-grade **AI support agent using LangGraph**, enabling **support team of 10–15 members** to query and update live database records via natural language — processing requests raised by clients and end users without any SQL expertise required.
- **Reduced query turnaround from 10–15 minutes of manual SQL writing to under 30 seconds end-to-end, including query generation, review, and user-approved execution.**
- Implemented **multi-layer guardrails** including malicious and off-topic query detection alongside **role-based access controls (RBAC)**, ensuring the agent could only perform actions within each user's authorized scope.
- Integrated **persistent agentic memory** to maintain context across sessions, enabling natural multi-turn conversations and eliminating repetitive re-prompting by users.
- Engineered **tool dispatch using Model Context Protocol (MCP)**, enabling schema-aware, dynamic database operations that adapt to user intent and session context — no hardcoded logic.
- **Partnered directly with the support team** to define scope — identifying relevant tables, columns, and use cases — then iterated on the system through structured UAT feedback cycles, delivering a solution validated and adopted by end users.

LLM-Powered Analytics Platform (NL -> SQL Automation)

- Led end-to-end design and delivery of the organization's **first production LLM-powered analytics platform**, enabling **internal analyst teams** to query live databases in plain English — **eliminating SQL dependency** across business analyst workflows.
- **Iterated on the platform through client demo cycles, incorporating feedback directly into product improvements and roadmap prioritization.**
- Architected a **Retrieval-Augmented Generation (RAG) layer over database schemas**, enriching each column with semantic descriptions, real-world value contexts, and data types to drive accurate table selection across schemas with **80+ columns per table**.
- Engineered **multi-table JOIN reasoning** by embedding table-level descriptions with primary/foreign key relationships, enabling the system to autonomously determine when cross-table joins were required and generate correct multi-source queries.
- Built a **full query execution pipeline** with dynamic DataFrame generation and visualization outputs, reducing analyst query turnaround from hours of manual effort to near real-time self-serve.
- Built a **custom LLM evaluation tracker** across 4 models (**LLaMA 3.1 8B, LLaMA 3.2 3B, GPT-3.5, GPT-OSS 20B**) — tracked **SQL generation accuracy, prompt versions, and per-model regressions** using a structured eval log; drove final model selection through iterative quality improvement cycles.
- Collaborated with a **3-person BA team** to systematically document table semantics, directly improving **model grounding and query reliability**; presented platform roadmap to senior management, **securing organizational buy-in**.

Business Rule Engine (BRE) for Loan Policy Intelligence

- Designed and delivered the organization's first AI-driven BRE, parsing complex loan and credit policy documents into structured, executable logical expressions — **used daily by COE and credit teams, with 4 policies spanning ~80 distinct business rules processed and validated since deployment.**
- Fine-tuned **Qwen 2.5 Coder 7B using LoRA via the Unsloth framework** on a custom domain dataset comprising business rules paired with their expected structured-logic outputs; improved **rule extraction accuracy from 78% to 90%** (evaluated on a held-out set of 15 unseen policies spanning varying complexity levels).
- Previously, BA/COE analysts manually read each policy and hand-coded logic expressions — a labor-intensive, error-prone process. Post-deployment, **human involvement is limited to verifying AI output and minor tweaks**, validated by the BA team across 5–10 policies of varying complexity.
- Built a **natural-language chatbot interface** enabling credit teams to query, explain, and dynamically update business rules — removing the need for technical intervention on routine rule modifications.
- Reduced loan policy review turnaround **from ~1 week to under 4 hours — a 95%+ reduction in cycle time** — confirmed through structured BA-led UAT across policies of varying length and complexity.

Credit Risk Modelling

- Engineered and optimized **ensemble classification models (Random Forest, LightGBM)** for loan approval workflows, improving **ROC-AUC on held-out test data from 0.64 to 0.80** through systematic hyperparameter tuning across model families.
- Built **scalable preprocessing pipelines** for large-scale financial datasets handling missing values, feature encoding, and class imbalance; conducted financial attribute evaluation to identify high-signal predictors of credit default.

Financial Document Intelligence Pipeline

- Designed and shipped a production document intelligence pipeline **processing 10,000+ CIBIL credit bureau PDFs/month** — **reducing annotation team manual extraction effort by 60%** compared to the prior fully manual per-document workflow.
- Engineered a **hybrid extraction architecture** combining **PyMuPDF for text extraction** and **pdfplumber for table parsing**; a single **300-page bureau document processes in under 30 seconds**. Integrated **LangChain-orchestrated LLM calls via OpenAI API** to generate **semantic commentary and contextual summaries**, transforming raw credit data into interpretable, analyst-ready JSON output.
- Output directly consumed by the Analytics team for downstream feature extraction used in credit risk model building — eliminating a manual data handoff between document processing and model development

SKILLS

Languages	Python · SQL · JavaScript · Java · C++
LLM / Agents	LangChain · LangGraph · CrewAI · AutoGen · Model Context Protocol (MCP) · RAG · Agentic AI · Prompt Engineering
ML / DL	TensorFlow · JAX · PyTorch · Hugging Face Transformers · Scikit-learn · LightGBM · NLP · Vector DBs (Pinecone · Weaviate)
Fine-Tuning	LoRA / PEFT · Unsloth · LLaMA · Model Evaluation
Data / Infra	PyMuPDF · pdfplumber · OpenAI API · Git · Pandas · NumPy · Docker · GCP
Soft Skills	Cross-Functional Collaboration · Stakeholder Communication · UAT & Feedback Cycles · Technical Presentation

EDUCATION

Bachelor’s Degree in Computer Science and Engineering2020-2024

Mahindra University | Hyderabad, India

CERTIFICATIONS

- **IBM AI Developer** — IBM (Feb 2026)
- **Fundamentals of MCP** — Hugging Face (Oct 2025)
- **Enterprise Risk Management – Level 1** — Institute of Risk Management (Mar 2024)